

PROFESSIONAL PORTFOLIO | AI Customer Review Analyzer

Mona Khaled

Python Developer | Data Scraping | AI Review Analytics | Business Insights

PROJECT TITLE

AI-Powered Customer Feedback Intelligence System

A professional end-to-end system designed to collect customer reviews from e-commerce platforms and transform them into **actionable business insights** using Python, AI analysis, and data visualization.

EXECUTIVE SUMMARY

This project was built to solve a real business problem: companies receive thousands of customer reviews that are difficult to analyze manually.

The system automates the full workflow from: **data extraction → text cleaning → AI analysis → visualization → exportable reports**.

It helps businesses quickly understand customer sentiment, recurring complaints, and improvement opportunities.

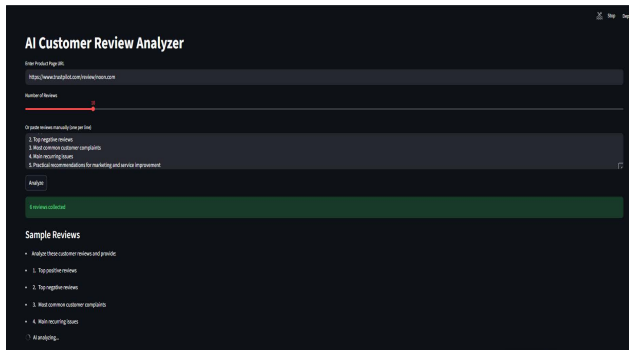
KEY FEATURES

- Smart review scraping from product/company pages
 - Manual review input support
 - Positive & negative feedback analysis
 - Common complaints detection
 - Recurring issue extraction
 - Practical business recommendations
 - Data visualization charts
 - CSV export
 - PDF report export
 - User-friendly web interface
-

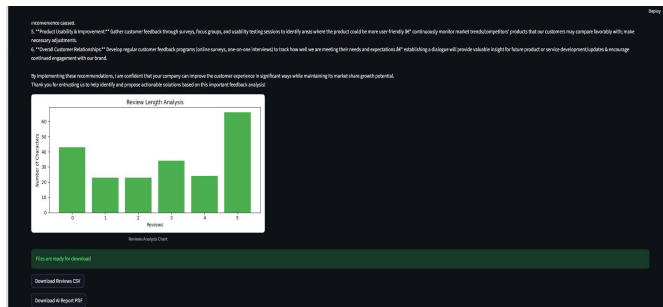
(SYSTEM INTERFACE (PROJECT SCREENSHOTS

Main Dashboard

Figure 1: Main Interface – User dashboard for inputting product URL, selecting number of reviews, and initiating AI analysis



AI Analysis Output + Chart Visualization



TECHNICAL STACK

- Python
- Pandas
- BeautifulSoup / Selenium
- AI-based NLP Analysis
- Matplotlib
- PDF & CSV Export
- Web UI Framework

WORKFLOW

1. Data Collection

Collects customer reviews from online review pages.

2. Data Processing

Cleans text, removes duplicates, and prepares structured data.

3. AI Analysis

Extracts: - top positive reviews - top negative reviews - recurring issues - business recommendations

4. Visualization

Creates charts and visual insights.

5. Reporting

Exports business-ready CSV and PDF reports.

BUSINESS VALUE

This project supports companies in: - customer satisfaction analysis - marketing optimization - product improvement - service quality enhancement - competitor benchmarking

MY ROLE

- Designed the project architecture
 - Developed the scraping engine
 - Built AI analysis workflow
 - Designed interface and user flow
 - Created visualization and reports
 - Converted results into business insights
-

PROFESSIONAL PROJECT STATEMENT

Built a production-ready AI-powered customer review analytics system that converts raw customer feedback into clear business decisions and actionable strategic recommendations.
